

1 Solution — GIAM 7.4

Problem 2

1.1 Problem

Find $(2x + 3)^5$.

1.2 Setup — Binomial Theorem

The binomial theorem states

$$(u + v)^n = \sum_{k=0}^n \binom{n}{k} u^{n-k} v^k.$$

With $u = 2x$, $v = 3$, and $n = 5$,

$$(2x + 3)^5 = \sum_{k=0}^5 \binom{5}{k} (2x)^{5-k} 3^k.$$

Each term factors into three pieces:

1. **Binomial coefficient** $\binom{5}{k}$ — from Pascal's row 5.
2. **Power of $2x$:** $(2x)^{5-k} = 2^{5-k} x^{5-k}$.
3. **Power of 3:** 3^k .

So the coefficient on x^{5-k} is $\binom{5}{k} \cdot 2^{5-k} \cdot 3^k$.

1.3 Step 1 — Pascal's Row 5

$$\binom{5}{0} = 1, \quad \binom{5}{1} = 5, \quad \binom{5}{2} = 10, \quad \binom{5}{3} = 10, \quad \binom{5}{4} = 5, \quad \binom{5}{5} = 1.$$

(*Spot check:* $1 + 5 + 10 + 10 + 5 + 1 = 32 = 2^5$, consistent with $(1 + 1)^5$.)

1.4 Step 2 — Compute Each Term

Lay out the three factors per term:

k	$\binom{5}{k}$	2^{5-k}	3^k	Coefficient $\binom{5}{k} \cdot 2^{5-k} \cdot 3^k$	Monomial
0	1	32	1	$1 \cdot 32 \cdot 1 = 32$	x^5
1	5	16	3	$5 \cdot 16 \cdot 3 = 240$	x^4
2	10	8	9	$10 \cdot 8 \cdot 9 = 720$	x^3
3	10	4	27	$10 \cdot 4 \cdot 27 = 1080$	x^2
4	5	2	81	$5 \cdot 2 \cdot 81 = 810$	x
5	1	1	243	$1 \cdot 1 \cdot 243 = 243$	1

Table 1.1

1.5 Step 3 — Assemble the Polynomial

$$(2x + 3)^5 = 32x^5 + 240x^4 + 720x^3 + 1080x^2 + 810x + 243.$$

1.6 Sanity Checks

$x = 0$. $(0 + 3)^5 = 243$. Polynomial at $x = 0$: 243. ✓

$x = 1$. $(2 + 3)^5 = 5^5 = 3125$. Polynomial at $x = 1$:

$$32 + 240 + 720 + 1080 + 810 + 243 = 3125. \checkmark$$

$x = -\frac{3}{2}$. $(2 \cdot (-\frac{3}{2}) + 3)^5 = 0^5 = 0$. So the polynomial must have $x = -\frac{3}{2}$ as a root.
Evaluating:

$$32(-\frac{3}{2})^5 + 240(-\frac{3}{2})^4 + 720(-\frac{3}{2})^3 + 1080(-\frac{3}{2})^2 + 810(-\frac{3}{2}) + 243$$

$$= -243 + 1215 - 2430 + 2430 - 1215 + 243 = 0. \checkmark$$

1.7 Remark — Where the Peak Coefficient Lives

Reading off the table, the coefficients **32, 240, 720, 1080, 810, 243** rise then fall, peaking at **1080** (the x^2 coefficient). For a plain binomial like $(x+1)^5$, the peak sits at the *middle* term (the symmetric peak of Pascal's row). Here the peak has *shifted* toward the lower-degree side because the powers of **3** grow faster than the powers of **2**.

The shift is captured by the ratio of consecutive coefficients. If $T_k = \binom{5}{k} \cdot 2^{5-k} \cdot 3^k$ is the coefficient at index k ,

$$\frac{T_{k+1}}{T_k} = \frac{\binom{5}{k+1}}{\binom{5}{k}} \cdot \frac{3}{2} = \frac{5-k}{k+1} \cdot \frac{3}{2}.$$

k	T_k	T_{k+1}/T_k	Direction
0	32	$5 \cdot \frac{3}{2} = 7.5$	rising sharply
1	240	$2 \cdot \frac{3}{2} = 3.0$	rising
2	720	$1 \cdot \frac{3}{2} = 1.5$	still rising
3	1080	$\frac{1}{2} \cdot \frac{3}{2} = 0.75$	falling
4	810	$\frac{1}{5} \cdot \frac{3}{2} = 0.30$	falling sharply
5	243	—	(last term)

Table 1.2

The ratio crosses 1 between $k=2$ and $k=3$ — solving $\frac{(5-k) \cdot 3}{(k+1) \cdot 2} = 1$ gives $k = \frac{13}{5} = 2.6$. So the peak coefficient lives at $k=3$ (the term above the crossing), which is $T_3 = 1080$, the x^2 coefficient. ✓

In general, for $(ax+b)^n$, the peak coefficient is at the value of k where $\frac{a(n-k)}{b(k+1)}$ transitions from > 1 to < 1 — equivalently, $k \approx \frac{an-b}{a+b}$.

1.8 Remark — A Useful Mnemonic

Each row of the binomial-theorem table reads " $\binom{n}{k} \cdot (\text{first}^{n-k}) \cdot (\text{second}^k)$." A common student error is to forget to raise the *coefficient* inside the binomial. For example, writing $(2x)^{5-k}$ as $2x^{5-k}$ instead of $2^{5-k}x^{5-k}$. The columns " 2^{5-k} " and " 3^k " in the table above are there specifically to keep these factors visible and hard to miss.

A second common error is to swap the exponents of the two terms. The rule is: *the first binomial term gets the descending exponent $(n, n-1, \dots, 0)$, and the second gets the ascending one.* In $(2x + 3)^5$, that means $2x$ has exponents $5, 4, 3, 2, 1, 0$ across the six terms, while 3 has exponents $0, 1, 2, 3, 4, 5$. The total exponent in each term is always $5 = n$. ✓

1.9 Remark — Decomposing Into “Pure Pascal”

A more compact way to write the answer:

$$(2x + 3)^5 = \sum_{k=0}^5 \binom{5}{k} 2^{5-k} 3^k x^{5-k}.$$

The coefficients $\{32, 240, 720, 1080, 810, 243\}$ are the pointwise product of:

Row	$k = 0$	$k = 1$	$k = 2$	$k = 3$	$k = 4$	$k = 5$
Pascal's row 5 — $\binom{5}{k}$	1	5	10	10	5	1
Powers of 2 — 2^{5-k}	32	16	8	4	2	1
Powers of 3 — 3^k	1	3	9	27	81	243
Product (coefficient on x^{5-k})	32	240	720	1080	810	243

Table 1.3

This decomposition makes the structure transparent: Pascal contributes the *symmetric* part, the powers of 2 favor *high-degree* terms, and the powers of 3 favor *low-degree* terms.